



## Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

### STUDENT CASE STUDY EXECUTION REPORT

|                |                          |
|----------------|--------------------------|
| Department     | FED                      |
| Subject        | DSA-2                    |
| Academic Year  | I-III Semester (2025-26) |
| Faculty Name   | Swetha Nagasri           |
| Student Name   | V Sameer Reddy           |
| Roll Number    | 2520030276               |
| Branch         | CSE                      |
| Course Outcome | CO3                      |

**Case Study Topic:** Tata Steel Pipeline Network using Edmonds-Karp Maximum Flow Algorithm

#### Work Done Summary:

In this case study, we analyzed a simplified Tata Steel pipeline network represented as a flow network. Each pipe has a maximum carrying capacity measured in kiloliters per hour. The objective was to determine the maximum possible flow from the source refinery (S) to the destination junction (T) using the Edmonds-Karp algorithm.

The algorithm repeatedly applies Breadth First Search (BFS) to identify the shortest augmenting path, computes the bottleneck capacity, updates residual capacities, and continues until no augmenting path remains.

#### Implementation Details:

- Modeled the pipeline network as a directed graph.
- Applied BFS to find augmenting paths from source S to sink T.
- Determined the bottleneck capacity for each path.
- Added bottleneck flow to the total flow.
- Updated residual capacities after every iteration.
- Repeated the process until no augmenting path existed.
- Calculated the maximum flow through the network.

#### Result / Output:

##### Edmonds-Karp Algorithm Steps

1. Use BFS to find an augmenting path from S to T.

2. Find the bottleneck capacity (minimum edge capacity on the path).
  3. Push flow equal to the bottleneck value.
  4. Update residual capacities.
  5. Repeat until no augmenting path exists.
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#### **Iteration 1**

Path:

$S \rightarrow A \rightarrow C \rightarrow T$

Bottleneck:

$\min(10, 4, 10) = 4$

Flow Added:

4

Total Flow:

4

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#### **Iteration 2**

Path:

$S \rightarrow A \rightarrow D \rightarrow T$

Remaining capacities:

$S \rightarrow A = 6$

$A \rightarrow D = 8$

$D \rightarrow T = 10$

Bottleneck:

$\min(6, 8, 10) = 6$

Flow Added:

6

Total Flow:

10

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#### **Iteration 3**

Path:

$S \rightarrow B \rightarrow C \rightarrow T$

Bottleneck:

$\min(10, 9, 6) = 6$

Flow Added:

6

Total Flow:

16

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#### **Iteration 4**

Path:

$S \rightarrow B \rightarrow C \rightarrow D \rightarrow T$

Remaining capacities:

4, 3, 6, 4

Bottleneck:

$\min(4, 3, 6, 4) = 3$

Flow Added:

3

Total Flow:

19

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**Maximum Flow**

Maximum Flow = 19 KL/hour

**Time Complexity****Edmonds-Karp Complexity:**

$$O(VE^2)$$

**Reason:**

- Each BFS traversal takes  **$O(V + E)$** .
- BFS always finds the shortest augmenting path.
- Number of augmentations is bounded by  **$O(VE)$** .
- Therefore total complexity becomes  **$O(VE^2)$** .
- BFS-based path selection guarantees polynomial-time execution.

**Conclusion:**

The Edmonds-Karp algorithm successfully computed the maximum flow through the Tata Steel pipeline network. By repeatedly finding shortest augmenting paths and updating residual capacities, the network achieved a **maximum flow of 19 KL/hour** from source **S** to destination **T**. The algorithm is efficient and guarantees polynomial-time performance for flow network problems.

**Sample Output:**

Maximum Flow = 19 KL/hour

GitHub Link: <https://github.com/sameerreddyvemireddy/DSA-Case-STUDIES>

| Student Signature | Faculty Signature |
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